Saddle Order and Aligned Training Dynamics in Tree Tensor Networks

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Abstract

We study gradient-flow training of tree tensor networks for targets that are finite sums of orthogonal product-basis tensors. The local geometry is governed by two distinct mechanisms. A missing component may be created directly by a connected perturbation along a subtree, or several cheaper partial perturbations may cooperate through a spectral instability. For a saddle representing one learned component we prove the exact formula

$$\operatorname{ord}(\theta_S) = \min\{k^\star, 2\ell^\star\},$$

where $k^\star$ is the smallest relevant Steiner-subtree size and $\ell^\star$ is the first level at which the residual coupling operator becomes unstable. The proof uses an orthogonal sector decomposition, a finite cumulant transform, a subtree-divisibility lemma, and matching direct and spectral constructions.

We then give an exact analogue of aligned deep-linear dynamics. On any routing-closed coordinate manifold the target amplitudes satisfy

$$\dot{y} = JJ^{\mathsf{T}}(s - y).$$

For product-basis supports satisfying an edgewise star condition, we construct such a manifold explicitly and prove that it is invariant. Its dynamics reduce to one scalar auxiliary equation per target component, with coupling determined exactly by the local entries shared by different components. Two components share the entry at a vertex precisely when that vertex lies outside the Steiner subtree of their disagreements. For a learned set J and a missing component r, the vertices at which r cannot reuse any learned entry form the connected routing region

$$\mathcal{R}_J(r) = \bigcap_{a \in J} \mathrm{St}(V_{ra}).$$

We prove that the corresponding canonical partial-target point is critical exactly when every such region has at least two vertices, that its full-space saddle order is at most $\min_r |\mathcal{R}_J(r)|$, and that its order within the invariant manifold is exactly this value. In the decoupled case the reduced equations become the Saxe–McClelland–Ganguli scalar mode equations, with architectural depth replaced by a state-dependent route depth.

The main text contains only proved statements. A conditional extension to arbitrary aligned later saddles, including a generalized eigenvalue criterion and the conjectural formula $\operatorname{ord} = \min\{d^\star, 2\ell^\star\}$, is placed in an appendix together with the precise missing connectedness hypothesis.

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1 Introduction

A tree tensor network (TTN) is a multilinear parameterization of a high-order tensor in which local tensors are contracted according to a tree. The absence of cycles makes the representation tractable, but it does not make its optimization geometry simple. A perturbation can contribute to the output only when its bond labels form a compatible routing through the tree, and the degree at which a new output component first appears depends on how many local entries must be changed simultaneously.

This paper studies that geometry for the squared-loss problem

$$L(\theta) = \frac{1}{2} \|W(\theta) - W^\star\|^2, \tag{1.1}$$

when the target is a finite sum of mutually orthogonal product-basis tensors. We are interested in parameter points whose contracted tensor represents only part of the target. Such points arise naturally in the small-initialization picture suggested by deep linear networks: training may remain for a long time near a configuration of low effective rank, then rapidly acquire a new component and move to another plateau. The local question is therefore:

> At what power of the parameter displacement does the first loss-decreasing perturbation appear, and what geometric mechanism produces it?

The answer already has two parts at a saddle representing one learned component. A missing component can be produced directly by opening a new bond channel along the smallest subtree connecting the cores on which it differs from the learned component. This is a connected route, and its order is the size of that Steiner subtree. There is also a genuinely different possibility: two cheaper partial changes can be individually unfavorable but jointly correlate with the residual. Their competition with the positive quadratic cost is governed by a finite coupling operator. The first spectral instability may occur before the direct route is available.

Our first main result gives the exact local order in this setting. If $k^\star$ is the smallest relevant Steiner-subtree size and $\ell^\star$ is the first scale at which the partial-change coupling operator has eigenvalue larger than the learned coefficient, then

$$\operatorname{ord}(\theta_S) = \min\{k^\star, 2\ell^\star\}. \tag{1.2}$$

The proof is not a formal Taylor expansion. Raw output coefficients contain products of independent changes on disjoint vertex sets, so their apparent degree does not reveal the degree of a genuinely connected perturbation. We remove these disconnected products by a finite cumulant transform. The resulting connected coordinate indexed by a set B is divisible by the unused-channel variables at every vertex of $\mathrm{St}(B)$. This converts tree connectivity into a sharp Taylor-order estimate. A filtered quadratic argument then shows that below the first spectral threshold no perturbation can improve on the direct or pairwise mechanisms.

The second part of the paper concerns dynamics. Saxe, McClelland, and Ganguli derived exact scalar mode equations for aligned deep linear networks [1]. Their modes are independent singular directions, and the depth of each mode is fixed by the architecture. TTNs admit an analogous reduction, but the natural objects are compatible routes rather than singular modes. On a routing-closed coordinate manifold with output amplitudes $y = (y_a)$, Euclidean gradient flow is exactly

$$\dot{x} = J(x)^{\mathsf{T}}(s - y(x)), \qquad \dot{y} = J(x)J(x)^{\mathsf{T}}(s - y), \tag{1.3}$$

where $J = Dy$. If an isolated route has d currently small factors and fixed active prefactor γ, balanced dynamics satisfy

$$\dot{y} = d\,\gamma^{2/d} y^{2-2/d}(s - y). \tag{1.4}$$

This is precisely the scalar deep-linear mode equation with depth d. In a TTN, however, d is not fixed: previously learned components may supply some of the required local entries, leaving only the remaining dormant factors to be learned.

An abstract invariant-manifold theorem does not by itself identify targets for which such a reduction is available. We therefore introduce a concrete support condition. For every tree edge, form the bipartite graph between the distinct target substrings on the two sides of the induced cut. We assume that every connected component of this graph is a star. Equivalently, the target support contains no component obtained by taking one side of one target string and the other side of a second target string across a single edge cut. Under this *edgewise star condition*, one bond channel may represent an entire rank-one support block, and components may share local entries without producing unintended recombinant strings.

For this target class we prove four structural facts.

1. The block-aligned TTN manifold is invariant under gradient flow.
2. Two target components share the local entry at a vertex v if and only if v lies outside the Steiner subtree of their disagreement set.
3. The exact dynamics reduce to one scalar auxiliary equation per target component. The components are coupled only through the entries that they share.
4. If J is the learned set and r is missing, then the vertices where r cannot reuse any learned local entry form the connected subtree

$$\mathcal{R}_J(r) = \bigcap_{a \in J} \mathrm{St}(V_{ra}). \tag{1.5}$$

The last identity gives a closed form for the state-dependent connected route depth. It includes both ordinary budding from one learned component and genuine branchwise splicing of several learned components. At the canonical partial-target point θ_J, put

$$c_J^\star := \min_{r \notin J} |\mathcal{R}_J(r)|\,. \tag{1.6}$$

We prove that θ_J is critical exactly when $c_J^\star \geq 2$, that

$$\mathrm{ord}(\theta_J) \leq c_J^\star \tag{1.7}$$

in the full parameter space, and that the order restricted to the invariant block-aligned manifold is exactly $c_J^\star$. The inequality can be strict in the full space because transverse partial changes may cooperate spectrally, just as they do at the one-component saddle.

These results give a clean distinction between what is known and what remains open. The complete one-component order formula is a full-space theorem. The edgewise star condition gives exact multi-component dynamics and an exact connected route order on an invariant manifold, together with a full-space criticality criterion and direct upper bound. A second exact later-saddle theorem is available under a stronger block-separation assumption that removes cross-parent interactions. For an arbitrary aligned multi-channel saddle, the expected formula is again the minimum of a connected direct order and twice a spectral filtration level, but a multi-channel analogue of the connected-coordinate divisibility lemma is still missing. We state that conditional result only in section B.

The organization is as follows. Section 2 introduces the TTN model and the local order. Sections 3 and 4 prove the exact one-component theorem, and section 5 records its local dynamical consequences. Section 6 gives the general route-amplitude reduction and conservation laws. Section 7 constructs the invariant block-aligned manifold under the edgewise star condition, and section 8 analyzes its later saddles. Section 9 gives the complementary block-separated full-space theorem. The appendices contain technical bookkeeping, the conditional unrestricted theory, and formal small-initialization consequences that are not claimed as a complete saddle-to-saddle theorem.

2 Model, target, and local saddle order

2.1 Tree tensor networks

Let $T = (V, E)$ be a finite tree with $K = |V|$ vertices. Each vertex v carries a physical space

$$\mathcal{H}_v = \mathbb{R}^{D_v}, \tag{2.1}$$

and each edge e has bond dimension χ_e. A core tensor at v is

$$G_v \in \mathcal{H}_v \otimes \bigotimes_{e \ni v} \mathbb{R}^{\chi_e}. \tag{2.2}$$

Contracting every pair of bond legs gives

$$W(\theta) \in \mathcal{H} := \bigotimes_{v \in V} \mathcal{H}_v, \tag{2.3}$$

where $\theta = (G_v)_{v \in V}$ is the full parameter vector. We equip parameter space with the Euclidean norm induced by the Frobenius norms of the cores.

Fix a basis in each bond space. A contraction monomial is specified by assigning a bond label to every edge and then choosing, at each vertex, a core entry whose incident bond labels agree with that assignment. This elementary observation is the source of the routing interpretation used below.

2.2 Orthogonal product-basis targets

The target tensor is

$$W^\star = \sum_{a \in \mathcal{N}} s_a U_a, \qquad s_a > 0, \tag{2.4}$$

where $\mathcal{N}$ is finite and

$$U_a = \bigotimes_{v \in V} q_{a,v} \tag{2.5}$$

with each $q_{a,v}$ a standard basis vector of $\mathcal{H}_v$. We assume that the strings are distinct, so the tensors U_a are mutually orthonormal:

$$\langle U_a, U_b \rangle = \delta_{ab}. \tag{2.6}$$

The loss is

$$L(\theta) = \frac{1}{2} \|W(\theta) - W^\star\|^2 . \tag{2.7}$$

This is the population squared loss after whitening; no statistical argument is used below beyond the Euclidean form (2.7).

It is often convenient to identify $q_{a,v}$ with its basis index $x_a(v)$. For two components a, b, define their disagreement set

$$V_{ab} := \{v \in V : x_a(v) \neq x_b(v)\} . \tag{2.8}$$

For $B \subseteq V$, let $\mathrm{St}(B)$ be the smallest connected subtree containing B. We set $\mathrm{St}(\varnothing) = \varnothing$.

2.3 Gauge transformations

Let $e = (u, w)$ be an edge. Matricize the two incident cores so that the e-leg is the column index. For any $A \in \mathrm{GL}(\chi_e)$, the transformation

$$G_{u;e} \longmapsto G_{u;e} A, \qquad G_{w;e} \longmapsto G_{w;e} A^{-\mathsf{T}} \tag{2.9}$$

leaves $W(\theta)$ unchanged. Diagonal choices of A redistribute scalar amplitude along a bond. Because the tree has no cycles, a one-channel product component can be balanced across all vertices by such diagonal gauge transformations.

The order defined below is unchanged by a local C^1 reparameterization with nonsingular derivative. Such a map changes Euclidean distance only by multiplicative constants in a sufficiently small neighbourhood. We shall therefore choose balanced or block-adapted representatives whenever useful, while always stating the representative at which a local theorem is applied.

2.4 Local order

Let θ_S be a critical point and $L_S = L(\theta_S)$. Define

$$\psi(\tau) := \min_{\|\theta - \theta_S\| \leq \tau} \big(L(\theta) - L_S\big). \tag{2.10}$$

Since L is polynomial and the ball is semialgebraic, ψ is semialgebraic. If it is not identically zero near $\tau = 0$, there are $c > 0$ and a positive rational number p such that

$$\psi(\tau) = -c\tau^p(1 + o(1)). \tag{2.11}$$

Definition 2.1 (Local saddle order)**.** The exponent p in (2.11) is the *order* of θ_S, denoted $\mathrm{ord}(\theta_S)$.

Equivalently, p is the largest exponent for which a local lower bound

$$L(\theta) - L_S \geq -C \|\theta - \theta_S\|^p \tag{2.12}$$

holds. At a critical point, $p = 2$ exactly when the Hessian has a negative eigenvalue. Orders $p > 2$ describe degenerate saddles whose first downhill term occurs beyond quadratic order.

If $\mathcal{M}$ is a smooth or coordinate submanifold through θ_S, we similarly define the restricted order $\mathrm{ord}_{\mathcal{M}}(\theta_S)$ by minimizing over $\mathcal{M} \cap B(\theta_S, \tau)$. Since the restricted minimization is over a smaller set,

$$\mathrm{ord}(\theta_S) \leq \mathrm{ord}_{\mathcal{M}}(\theta_S) \tag{2.13}$$

whenever both orders are defined.

3 The one-component saddle

Fix $j \in \mathcal{N}$ and abbreviate

$$q_v := q_{j,v}, \qquad U_j = \bigotimes_{v \in V} q_v. \tag{3.1}$$

We study a parameter point θ_S with

$$W(\theta_S) = s_j U_j. \tag{3.2}$$

Choose a representative in which every core uses one distinguished bond channel, labelled 0, and all other channels are inactive:

$$G_v^S = a_v q_v \otimes \bigotimes_{e \ni v} e_{0,e}, \qquad \prod_{v \in V} a_v = s_j. \tag{3.3}$$

By diagonal gauge transformations we may take

$$a_v = a := s_j^{1/K} \tag{3.4}$$

for every v.

The residual and saddle loss are

$$R := W^\star - s_j U_j = \sum_{r \neq j} s_r U_r, \qquad L_S = \frac{1}{2} \|R\|^2 . \tag{3.5}$$

For each residual component let

$$V_r := V_{rj} = \{v : q_{r,v} \neq q_v\}, \qquad k_r := |\mathrm{St}(V_r)|, \qquad k^\star := \min_{r \neq j} k_r. \tag{3.6}$$

Thus $k^\star$ is the smallest number of vertices in a connected subtree that supports a missing component relative to U_j.

3.1 Orthogonal difference sectors

Define

$$p_v = q_v q_v^{\mathsf{T}}, \qquad p_v^\perp = \mathrm{I} - p_v. \tag{3.7}$$

For $B \subseteq V$, let

$$P_B := \bigotimes_{v \in B} p_v^\perp \otimes \bigotimes_{v \notin B} p_v. \tag{3.8}$$

These 2^K projections are mutually orthogonal and sum to the identity. Their ranges are naturally identified with

$$\mathcal{H}_B := \bigotimes_{v \in B} q_v^\perp, \tag{3.9}$$

with the factors q_v outside B understood. Write

$$R_B := P_B R. \tag{3.10}$$

For the product-basis target,

$$R_B = \sum_{r : V_r = B} s_r U_r, \qquad k^\star = \min \{|\mathrm{St}(B)| : R_B \neq 0\} . \tag{3.11}$$

Lemma 3.1 (Criticality)**.** *The point θ_S is critical if and only if*

$$R_{\{v\}} = 0 \tag{3.12}$$

for every $v \in V$. For the product-basis target, this is equivalent to $k^\star \geq 2$.

Proof. A first-order perturbation of a core entry on an unused bond channel produces no first-order output change, because the adjacent core has zero amplitude on the matching label. A first-order perturbation of the active entry at v changes the output only in the sectors $\mathcal{H}_{\varnothing}$ and $\mathcal{H}_{\{v\}}$. Since $R \perp U_j$, the residual has no component in $\mathcal{H}_{\varnothing}$. Therefore

$$dL(\delta\theta) = \langle W(\theta_S) - W^\star, dW(\delta\theta)\rangle = -\langle R, dW(\delta\theta)\rangle \tag{3.13}$$

vanishes for every perturbation if and only if every singleton residual sector is zero. A nonzero singleton sector is exactly a product component differing from U_j at one vertex. □

We assume from now on that $R \neq 0$ and $k^\star \geq 2$.

3.2 Partial changes and the spectral filtration

A nonempty set $B \subseteq V$ together with a unit product-basis tensor $\beta \in \mathcal{H}_B$ defines an *atom* $n = (B, \beta)$. Its cost is

$$h(n) := |\mathrm{St}(B)|\,. \tag{3.14}$$

A singleton atom is produced by rotating one physical vector within the active channel. A multi-vertex atom is produced by opening a private channel along its Steiner subtree.

For $\ell \geq 1$, let $\mathcal{N}_\ell$ be the finite set of atoms with $h(n) \leq \ell$. For $n_i = (B_i, \beta_i)$ define the symmetric nonnegative matrix

$$M_\ell[n_1, n_2] = \begin{cases} s_r, & B_1 \cap B_2 = \varnothing \text{ and the two atoms combine to } U_r, \\ 0, & \text{otherwise.} \end{cases} \tag{3.15}$$

Equivalently, if e_{n_i} is the corresponding unit basis tensor, then for disjoint B_1, B_2,

$$M_\ell[n_1, n_2] = \langle R_{B_1 \sqcup B_2}, e_{n_1} \otimes e_{n_2}\rangle\,. \tag{3.16}$$

Set

$$\mu_\ell := \lambda_{\max}(M_\ell), \qquad \ell^\star := \min\{\ell \geq 1 : \mu_\ell > s_j\}\,, \tag{3.17}$$

with $\ell^\star = \infty$ if the threshold is never crossed.

The direct and spectral scales $k^\star$ and $2\ell^\star$ are the only two scales that survive the lower-bound argument. The main theorem is the following.

Theorem 3.2 (One-component saddle order)**.** *Assume that*

1. $R \neq 0$ *and* $k^\star \geq 2$*;*
2. $\mu_\ell \neq s_j$ *for every* $\ell < k^\star/2$*;*
3. *enough unused bond channels are available for the direct and spectral constructions used below.*

Then

$$\boxed{\mathrm{ord}(\theta_S) = \min\{k^\star, 2\ell^\star\}.} \tag{3.18}$$

The proof occupies section 4. The nonboundary assumption excludes the codimension-one case in which positive quadratic cost and residual coupling cancel exactly; higher-order terms can then change the answer.

4 Proof of the one-component order theorem

4.1 The master loss identity

Define the learned-component amplitude

$$\omega(\theta) := \langle U_j, W(\theta) \rangle . \tag{4.1}$$

Since $\omega(\theta_S) = s_j > 0$, it is nonzero near θ_S. For $B \neq \varnothing$, put

$$m_B := \frac{P_B W}{\omega} \in \mathcal{H}_B, \qquad m_\varnothing := 1. \tag{4.2}$$

Then

$$W = \omega \left(U_j + \sum_{B \neq \varnothing} m_B \right), \tag{4.3}$$

with orthogonality across the sectors.

Proposition 4.1 (Master loss identity)**.** *For every nearby parameter point with $\omega \neq 0$,*

$$L - L_S = \frac{1}{2}(\omega - s_j)^2 + \sum_{B \neq \varnothing} \left[\frac{\omega^2}{2} \|m_B\|^2 - \omega \langle R_B, m_B \rangle \right]. \tag{4.4}$$

Proof. Orthogonality gives

$$P_\varnothing (W - W^\star) = (\omega - s_j) U_j \tag{4.5}$$

and, for $B \neq \varnothing$,

$$P_B (W - W^\star) = \omega m_B - R_B. \tag{4.6}$$

Subtracting $L_S = \frac{1}{2} \sum_{B \neq \varnothing} \|R_B\|^2$ and expanding proves (4.4). □

The first term and the terms $\frac{1}{2}\omega^2 \|m_B\|^2$ are nonnegative. A loss decrease must therefore come from residual correlations $\langle R_B, m_B \rangle$. The difficulty is that m_B contains both connected changes on B and products of independent changes on smaller disjoint sets.

4.2 Connected coordinates

For families $f = (f_B)_{B \subseteq V}$ and $g = (g_B)_{B \subseteq V}$ with $f_B, g_B \in \mathcal{H}_B$, define the partition convolution

$$(f * g)_B := \sum_{B_1 \sqcup B_2 = B} f_{B_1} \otimes g_{B_2}. \tag{4.7}$$

Because the tensor factors are labelled by vertices, this product is commutative and associative. A family supported on nonempty sets is nilpotent, so exponential and logarithm in this algebra are finite sums.

Since $m_\varnothing = 1$, define

$$\kappa := \log_* m. \tag{4.8}$$

The moment–cumulant identities are

$$m_B = \sum_{\pi \in \mathrm{Part}(B)} \bigotimes_{A \in \pi} \kappa_A, \tag{4.9}$$

and

$$\kappa_B = \sum_{\pi \in \mathrm{Part}(B)} (-1)^{|\pi|-1} (|\pi| - 1)! \bigotimes_{A \in \pi} m_A. \tag{4.10}$$

Thus κ_B is the connected part of the normalized output sector m_B.

For each vertex v, let ν_v be the vector of core entries that carry at least one nonzero bond label. These entries vanish at θ_S.

Lemma 4.2 (Subtree divisibility)**.** *If $|B| \geq 2$, then in a neighbourhood of θ_S,*

$$\|\kappa_B\| \leq C \prod_{x \in \mathrm{St}(B)} \|\nu_x\| . \tag{4.11}$$

In particular, if $\varepsilon = \|\theta - \theta_S\|$, then

$$\|\kappa_B\| \leq C\varepsilon^{|\,\mathrm{St}(B)|}. \tag{4.12}$$

Proof. Fix $x \in \mathrm{St}(B)$ and set $\nu_x = 0$. Then every nonzero entry of core x uses bond label 0 on every incident edge. Removing x separates the tree into branches V_b, and the contraction factorizes as

$$W = g_x \otimes \bigotimes_b W_b. \tag{4.13}$$

The overlap ω factorizes in the same way. After normalization, the sector family factorizes in the convolution algebra:

$$m = m^{(x)} \mathbin{*\!*}_b m^{(b)}. \tag{4.14}$$

Taking $\log_*$ shows that κ is supported only on subsets of $\{x\}$ or on subsets contained in one branch. Since $x \in \mathrm{St}(B)$ and $|B| \geq 2$, the set B is of neither form. Hence $\kappa_B = 0$ whenever $\nu_x = 0$.

The map $\theta \mapsto \kappa_B$ is analytic. Its power series therefore contains, in every nonzero monomial, at least one coordinate from each block ν_x, $x \in \mathrm{St}(B)$. Factoring one such coordinate from every block and bounding the remaining convergent series on a smaller neighbourhood gives (4.11). □

This lemma is the structural core of the proof. A raw sector can be assembled from disconnected pieces without using every connector vertex. Its connected part cannot: if the network is forced through the active channel at any connector vertex, the normalized output factorizes there and the connected coordinate vanishes.

4.3 The direct connected construction

Choose r with $k_r = k^\star$ and put $S = \mathrm{St}(V_r)$. Give every edge of S one unused label, say 1. At each $v \in S$, add a core entry of size τ with physical vector $q_{r,v}$, label 1 on the edges of S incident to v, and label 0 on edges leaving S. Leave all other entries unchanged.

A contraction term that chooses the new entry at one endpoint of an internal edge and the old entry at the other vanishes. Since S is connected, only the all-old and all-new choices survive. Consequently

$$W = s_j U_j + y U_r, \qquad y = a^{K-k^\star}\tau^{k^\star}. \tag{4.15}$$

Therefore

$$L - L_S = -s_r y + \frac{1}{2}y^2 = -c\tau^{k^\star} + O(\tau^{2k^\star}), \tag{4.16}$$

and hence

$$\mathrm{ord}(\theta_S) \leq k^\star. \tag{4.17}$$

4.4 The spectral instability threshold

Suppose atoms in $\mathcal{N}_\ell$ are assigned normalized output amplitudes ηx_n with $\|x\| = 1$. No individual atom has residual overlap when $\ell < k^\star$. To second order in η, the positive deviation cost is

$$\frac{s_j^2}{2}\eta^2 \|x\|^2 , \tag{4.18}$$

whereas compatible pairs give residual reward

$$\frac{s_j}{2}\eta^2 x^{\mathsf{T}} M_\ell x. \tag{4.19}$$

The leading quadratic form in the atom amplitudes is therefore

$$\frac{s_j}{2}\eta^2 \left(s_j \|x\|^2 - x^{\mathsf{T}} M_\ell x \right). \tag{4.20}$$

It has a negative direction exactly when $\mu_\ell > s_j$. At $\ell = 1$, this is the ordinary negative-Hessian criterion for the block of one-core physical rotations.

4.5 Lower bound below the first spectral instability

For $m \geq 1$, define

$$\mathcal{H}_{\leq m} := \bigoplus_{\substack{A \neq \varnothing \\ |\operatorname{St}(A)| \leq m}} \mathcal{H}_A. \tag{4.21}$$

On this space define the symmetric bilinear form

$$Q(x, y) := \sum_{A_1 \cap A_2 = \varnothing} \langle R_{A_1 \sqcup A_2}, x_{A_1} \otimes y_{A_2} \rangle, \tag{4.22}$$

and let S_m be the associated self-adjoint operator. In the product-basis coordinates used to define (3.15), S_m is represented by M_m.

Proposition 4.3 (Filtered lower bound)**.** *Let*

$$2 \leq \ell \leq \left\lceil \frac{k^\star}{2} \right\rceil. \tag{4.23}$$

If

$$\lambda_{\max}(S_{\ell-1}) < s_j, \tag{4.24}$$

then, for θ sufficiently close to θ_S,

$$L(\theta) - L_S \geq -C \|\theta - \theta_S\|^{\min(k^\star, 2\ell)}. \tag{4.25}$$

Proof. Put $\varepsilon = \|\theta - \theta_S\|$ and call a nonempty set A *small* if $|\operatorname{St}(A)| \leq \ell - 1$, and *large* otherwise. Let κ_s collect the connected coordinates indexed by small sets. By lemma 4.2, every large κ_A is $O(\varepsilon^\ell)$. If A is small, then $R_A = 0$, because every nonzero residual sector has Steiner size at least $k^\star$.

For a small set B, every block of every partition of B is small. The moment–cumulant formula therefore gives

$$\sum_{B \neq \varnothing} \|m_B\|^2 \geq (1 - C\varepsilon) \|\kappa_s\|^2. \tag{4.26}$$

Expand each residual term $\langle R_B, m_B \rangle$ over partitions of B. A one-block term is $O(\varepsilon^{k^\star})$. Two small blocks give $\frac{1}{2} Q(\kappa_s, \kappa_s)$. A small–large product is controlled by Young's inequality, two large blocks are $O(\varepsilon^{2\ell})$, and every partition with at least three blocks either contains two large blocks or contains two small blocks and an additional $O(\varepsilon)$ factor. Thus, for every fixed $\gamma > 0$,

$$\sum_{B \neq \varnothing} \langle R_B, m_B \rangle \leq \frac{1}{2} \left(\lambda_{\max}(S_{\ell-1}) + M\gamma + C\varepsilon \right) \|\kappa_s\|^2 + C_\gamma \varepsilon^p, \tag{4.27}$$

where $p = \min(k^\star, 2\ell)$.

Insert (4.26) and (4.27) into the master identity. Since $\omega = s_j + O(\varepsilon)$ and the gap in (4.24) is strict, first choose γ and then ε sufficiently small so that the coefficient of $\|\kappa_s\|^2$ is positive. Discarding this positive term gives (4.25). The term-by-term bookkeeping is written explicitly in section A. □

4.6 The spectral construction

Suppose $2\ell < k^\star$ and a principal submatrix $M_\ell[X]$ has largest eigenvalue $\rho > s_j$. Let $x \geq 0$ be a unit Perron eigenvector. Assign a private unused channel to every multi-vertex atom in X along its Steiner subtree. Give atom n normalized output amplitude ηx_n. A singleton atom uses an active-channel physical rotation of size $O(\eta)$; a multi-vertex atom of cost $h_n \leq \ell$ uses entries of size $O((\eta x_n)^{1/h_n})$. Hence

$$\|\theta - \theta_S\| = O(\eta^{1/\ell}). \tag{4.28}$$

Private labels force each multi-vertex atom to be selected completely or not at all. Since $2\ell < k^\star$, no individual atom has residual overlap, and every nonzero second-order residual term is one of the compatible pairs recorded by $M_\ell[X]$. Therefore

$$L - L_S = \frac{s_j}{2}(s_j - \rho)\eta^2 + O(\eta^3). \tag{4.29}$$

Taking $\eta \asymp \tau^\ell$ gives

$$\operatorname{ord}(\theta_S) \leq 2\ell. \tag{4.30}$$

4.7 Completion of the proof

Proof of theorem 3.2. The direct construction gives (4.17). If $2\ell^\star < k^\star$, minimality and the nonboundary assumption imply $\mu_{\ell^\star-1} < s_j$. Applying proposition 4.3 with $\ell = \ell^\star$ gives the lower bound $\operatorname{ord}(\theta_S) \geq 2\ell^\star$, while the spectral construction gives the reverse inequality.

If $2\ell^\star \geq k^\star$, choose $\ell = \lceil k^\star/2 \rceil$. Then $\ell - 1 < k^\star/2 \leq \ell^\star$, so $\mu_{\ell-1} < s_j$ by the same nonboundary argument. Proposition 4.3 gives $\operatorname{ord}(\theta_S) \geq k^\star$, and the direct construction gives equality. □

Corollary 4.4 (One weaker residual component)**.** *Suppose*

$$W^\star = s_j U_j + s_r U_r, \qquad 0 < s_r < s_j. \tag{4.31}$$

Then $\mu_\ell \leq s_r < s_j$ for every ℓ, so

$$\operatorname{ord}(\theta_S) = |\operatorname{St}(V_{rj})|. \tag{4.32}$$

Proof. Every row of M_ℓ has at most one nonzero entry, equal to s_r, so its spectral radius is at most s_r. Hence $\ell^\star = \infty$ and theorem 3.2 applies. □

If $k^\star = 2$, the saddle is strict. If $k^\star \in \{3, 4\}$, an order smaller than $k^\star$ can only come from $\ell^\star = 1$, hence from a negative Hessian direction. A positive-semidefinite Hessian together with a strictly smaller higher-order spectral route first becomes possible at $k^\star \geq 5$.

Example 4.5 (Order four with a positive-semidefinite Hessian). Consider a path with six vertices, learned component 000000 of coefficient 1, and two residual components

$$110001, \qquad 110010, \tag{4.33}$$

with coefficient 0.8 each. Their Steiner sizes are 6 and 5, so $k^\star = 5$. No residual component is the union of two one-vertex changes, hence $M_1 = 0$ and the relevant Hessian block is positive semidefinite.

At level $\ell = 2$, take the two-vertex atom 11 on vertices $0, 1$ and the singleton atoms at vertices 4 and 5. On these three atoms,

$$M_2 = \begin{pmatrix} 0 & 0.8 & 0.8 \\ 0.8 & 0 & 0 \\ 0.8 & 0 & 0 \end{pmatrix}, \qquad \lambda_{\max}(M_2) = 0.8\sqrt{2} > 1. \tag{4.34}$$

Thus $\ell^\star = 2$ and

$$\operatorname{ord}(\theta_S) = 4. \tag{4.35}$$

Indeed, if u, w create the two-vertex atom, $y = uw$, and α, β are the two singleton rotations, then to leading order

$$L - L_S = \frac{1}{2}(y^2 + \alpha^2 + \beta^2) - 0.8y(\alpha + \beta). \tag{4.36}$$

Minimizing over α, β gives $\alpha = \beta = 0.8y$ and $L - L_S = -0.14y^2$. Since y is quadratic in the core displacement, the descent is quartic.

5 Local dynamics at the one-component saddle

The order theorem is static: it describes the best loss decrease in a small parameter ball. It nevertheless determines the natural time scale whenever the flow is reduced to a leading unstable coordinate.

5.1 Scalar normal forms

Suppose a one-dimensional reduced loss has the form

$$L - L_S = -\frac{c}{p}b^p + o(b^p), \qquad c > 0. \tag{5.1}$$

Then

$$\dot{b} = cb^{p-1} + o(b^{p-1}). \tag{5.2}$$

For $p = 2$, the escape time from $b(0) = \delta$ to a fixed small threshold is

$$t_{\text{esc}} = \frac{1}{c}\log\frac{1}{\delta} + O(1). \tag{5.3}$$

For $p > 2$,

$$t_{\text{esc}} = \frac{\delta^{-(p-2)}}{(p-2)c} + O(1). \tag{5.4}$$

These formulas are exact consequences of the stated scalar normal form. A local order theorem alone does not prove that a full trajectory reaches the saddle in a generic direction or that its motion is one-dimensional; those are separate dynamical questions.

5.2 Exact direct-route dynamics

Let r be a residual component with $S = \operatorname{St}(V_r)$ and $h = |S|$. Restrict to the private-channel direct construction from section 4. Let b_v be the new-channel entry at $v \in S$ and hold the active background fixed at the balanced saddle. The residual amplitude is

$$y = a^{K-h}\prod_{v\in S} b_v, \tag{5.5}$$

and the restricted loss is exactly

$$L - L_S = -s_r y + \frac{1}{2}y^2. \tag{5.6}$$

Gradient flow therefore satisfies

$$\dot{b}_v = (s_r - y)a^{K-h}\prod_{u\in S\setminus\{v\}} b_u. \tag{5.7}$$

The quantities $b_v^2 - b_u^2$ are conserved. Hence balanced initial data remain balanced. If $b_v = b$ for all $v \in S$, then

$$\dot{b} = a^{K-h}\left(s_r - a^{K-h}b^h\right)b^{h-1}. \tag{5.8}$$

Equivalently, since $y = a^{K-h}b^h$,

$$\dot{y} = h\, a^{2(K-h)/h} y^{2-2/h}(s_r - y). \tag{5.9}$$

The leading escape scale is therefore logarithmic when $h = 2$ and proportional to $\delta^{-(h-2)}$ when $h \geq 3$.

Equation (5.9) already has the form of an aligned depth-h deep-linear mode equation. The more general route formulation in section 6 shows that this is not special to a one-component saddle: h is replaced by the number of factors that are still small in the current learned state.

6 Exact dynamics on routing-closed manifolds

This section isolates the precise structural condition under which a TTN has exact Saxe-like amplitude dynamics. The theorem is elementary, but it is useful because it separates the dynamical reduction from the combinatorial problem of constructing an invariant manifold.

6.1 A general invariant-manifold theorem

Let $\mathcal{M}$ be a coordinate subspace of TTN parameter space with coordinates $x = (x_1, \ldots, x_N)$. Suppose that, on $\mathcal{M}$,

$$W(x) = \sum_{a \in \mathcal{N}} y_a(x) U_a. \tag{6.1}$$

Thus every nonzero contraction term on $\mathcal{M}$ lies in the target span. Let z denote any forbidden core coordinate transverse to $\mathcal{M}$.

Theorem 6.1 (Exact route-amplitude dynamics)**.** *Assume* (6.1) *and*

$$\left.\frac{\partial W}{\partial z}\right|_{\mathcal{M}} \perp \operatorname{span}\{U_a : a \in \mathcal{N}\} \tag{6.2}$$

for every forbidden coordinate z. Then $\mathcal{M}$ is invariant under gradient flow for the loss (2.7)*. If $J(x) = Dy(x)$, the exact dynamics on $\mathcal{M}$ are*

$$\dot{x} = J(x)^{\mathsf{T}}(s - y(x)), \tag{6.3}$$

$$\dot{y} = K(x)(s - y), \qquad K(x) := J(x)J(x)^{\mathsf{T}}. \tag{6.4}$$

Proof. For a forbidden coordinate,

$$\frac{\partial L}{\partial z} = \langle W - W^\star, \partial W/\partial z \rangle = 0 \tag{6.5}$$

on $\mathcal{M}$, so no transverse velocity is generated. On $\mathcal{M}$,

$$L(x) = \frac{1}{2}\left\|y(x) - s\right\|^2, \tag{6.6}$$

and the chain rule gives (6.3). Differentiating $y(x(t))$ gives (6.4). □

The positive-semidefinite matrix $K = JJ^{\mathsf{T}}$ is the route-overlap kernel. Its off-diagonal entries quantify parameter sharing between different target amplitudes. Deep-linear singular modes are independent in the usual aligned basis; TTN routes need not be.

6.2 Monomial routes

A particularly transparent case is

$$y_a(x) = \gamma_a \prod_{i=1}^{N} x_i^{A_{ai}}, \qquad A_{ai} \in \{0, 1\}, \tag{6.7}$$

where A is the route–entry incidence matrix and $\gamma_a > 0$ is a fixed prefactor. For positive coordinates,

$$\dot{x}_i = \sum_a A_{ai}(s_a - y_a)\frac{y_a}{x_i}. \tag{6.8}$$

The amplitude kernel is

$$K_{ab}(x) = y_a y_b \sum_i \frac{A_{ai}A_{bi}}{x_i^2}. \tag{6.9}$$

Thus two amplitudes interact exactly through the local factors used by both routes.

Proposition 6.2 (Route-balance invariants)**.** *For every $c \in \ker A$,*

$$\sum_{i=1}^{N} c_i x_i^2 \tag{6.10}$$

is constant along the flow (6.8)*.*

Proof. Using (6.8),

$$\frac{\mathrm{d}}{\mathrm{d}t}\sum_i c_i x_i^2 = 2\sum_i c_i x_i \dot{x}_i \tag{6.11}$$

$$= 2\sum_a (s_a - y_a) y_a \sum_i A_{ai} c_i = 0. \tag{6.12}$$

□

For one route, this contains the familiar invariants $x_i^2 - x_j^2$. Equal positive factors therefore remain equal.

6.3 The scalar route equation

Consider one isolated route with d variable factors and fixed prefactor γ:

$$y = \gamma \prod_{i=1}^{d} b_i. \tag{6.13}$$

If $b_1 = \cdots = b_d = b$, then

$$\dot{b} = \gamma(s - \gamma b^d) b^{d-1}. \tag{6.14}$$

Since $y = \gamma b^d$,

$$\boxed{\dot{y} = d\,\gamma^{2/d} y^{2-2/d}(s - y).} \tag{6.15}$$

For $\gamma = 1$, this is the scalar depth-d equation of aligned deep linear networks [1]. The distinction is conceptual: in a TTN, d counts the local factors that are still dormant along the route. Factors already activated by previously learned components are absorbed into γ.

Starting from $b(0) = \delta$ and stopping at a fixed small threshold gives

$$t_{\text{learn}} \sim \begin{cases} (\gamma s)^{-1} \log(1/\delta), & d = 2, \\ \big((d-2)\gamma s\big)^{-1} \delta^{-(d-2)}, & d \geq 3. \end{cases} \tag{6.16}$$

A route with $d = 1$ is not a saddle direction: it has a nonzero first derivative.

6.4 Bond Gram conservation

The preceding balance laws are route-specific. A gauge-theoretic conservation law holds for every TTN gradient flow.

Proposition 6.3 (Bond Gram balance)**.** *Let $e = (u, w)$ be an edge, and matricize G_u and G_w with the e-leg as the column index. Then*

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(G_{u;e}^{\mathsf{T}} G_{u;e} - G_{w;e}^{\mathsf{T}} G_{w;e}\right) = 0. \tag{6.17}$$

Proof. The loss is invariant under

$$G_{u;e} \mapsto G_{u;e} A, \qquad G_{w;e} \mapsto G_{w;e} A^{-\mathsf{T}}. \tag{6.18}$$

Differentiating at $A = \mathrm{I} + \epsilon X$ for arbitrary X gives

$$\nabla_u^{\mathsf{T}} G_u = G_w^{\mathsf{T}} \nabla_w. \tag{6.19}$$

Under $\dot{G} = -\nabla L$,

$$\frac{\mathrm{d}}{\mathrm{d}t}(G_u^{\mathsf{T}} G_u) = -(\nabla_u^{\mathsf{T}} G_u + G_u^{\mathsf{T}} \nabla_u) \tag{6.20}$$

$$= -(G_w^{\mathsf{T}} \nabla_w + \nabla_w^{\mathsf{T}} G_w) = \frac{\mathrm{d}}{\mathrm{d}t}(G_w^{\mathsf{T}} G_w). \tag{6.21}$$

□

The edgewise Gram difference is the TTN analogue of layer balance in matrix factorizations and deep linear networks. In the block-aligned setting of the next section, its diagonal entries become conservation laws for shared channel amplitudes.

7 Edgewise star supports and exact component dynamics

The route-amplitude theorem becomes useful only after an invariant routing-closed manifold has been identified. We now give a concrete target class for which this can be done exactly. The class allows substantial parameter sharing and is strictly more general than assigning one independent bond channel to every target component.

7.1 Edge support graphs

Fix an edge e. Removing e splits the vertex set into two connected sides,

$$V = A_e \sqcup B_e. \tag{7.1}$$

For a target string x_a, write $x_a|_{A_e}$ and $x_a|_{B_e}$ for its two restrictions. Define a bipartite graph Γ_e whose left vertices are the distinct A_e-substrings, whose right vertices are the distinct B_e-substrings, and whose edge labelled by a joins $x_a|_{A_e}$ to $x_a|_{B_e}$.

Assumption 7.1 (Edgewise star condition)**.** For every tree edge e, every connected component of Γ_e is a star, where a single edge is allowed as a degenerate star.

Equivalently, there are no three distinct target components a, b, c such that, for one of the two orientations of an edge cut,

$$x_b|_{A_e} = x_a|_{A_e}, \qquad x_b|_{B_e} = x_c|_{B_e}. \tag{7.2}$$

Indeed, (7.2) is exactly a path of three edges in Γ_e. A connected bipartite graph has no such path if and only if it is a star.

Let $\mathcal{B}_e$ be the set of star components of Γ_e, and let

$$\beta_e(a) \in \mathcal{B}_e \tag{7.3}$$

be the block containing the edge labelled by a.

Proposition 7.2 (Rank across an edge)**.** *Under assumption 7.1, the matricization of $W^\star$ across the cut $A_e|B_e$ has rank $|\mathcal{B}_e|$.*

Proof. Within one star block, all edges share either one left substring or one right substring. The corresponding submatrix of the target matricization is therefore rank one. Distinct blocks use disjoint sets of left and right basis vectors, so the ranks add. □

Thus one bond channel per block is both sufficient and minimal for representing the target across that edge.

7.2 Local types and the block-aligned manifold

For a component a and vertex v, define its local type

$$\tau_v(a) := (x_a(v), (\beta_e(a))_{e \ni v}) \,. \tag{7.4}$$

Let

$$\mathcal{T}_v := \{\tau_v(a) : a \in \mathcal{N}\} \tag{7.5}$$

be the set of distinct local types at v. Choose on every edge a bond basis indexed by $\mathcal{B}_e$ and, if the bond space is larger, complete it arbitrarily.

The *block-aligned manifold* $\mathcal{M}_\star$ is the coordinate subspace with cores

$$G_v = \sum_{\tau \in \mathcal{T}_v} g_{v,\tau}\, e_{x(\tau)} \otimes \bigotimes_{e \ni v} e_{\beta_e(\tau),e}, \tag{7.6}$$

all other core entries being zero. Components with the same local type use the same scalar parameter.

The next lemma identifies exactly when this sharing occurs.

Lemma 7.3 (Local sharing and Steiner geometry)**.** *Under assumption 7.1, for any target components a, b and any vertex v,*

$$\tau_v(a) = \tau_v(b) \quad \Longleftrightarrow \quad v \notin \mathrm{St}(V_{ab}). \tag{7.7}$$

Proof. Suppose first that $v \notin \mathrm{St}(V_{ab})$. All disagreements between a and b lie in one branch of $T \setminus \{v\}$. Hence $x_a(v) = x_b(v)$. Across each edge incident to v, the two strings agree on at least one whole side of the cut, so their labelled edges in Γ_e share a vertex and belong to the same block. Thus their local types agree.

Conversely, suppose $v \in \mathrm{St}(V_{ab})$. If $x_a(v) \neq x_b(v)$, the local types differ immediately. Otherwise, the disagreement set meets at least two branches at v. Choose an incident edge leading into one of them. The strings differ on both sides of this edge cut. Two edges in the same star block must share the centre vertex and hence agree on one side. Therefore $\beta_e(a) \neq \beta_e(b)$, so the local types differ. □

Corollary 7.4 (Helly intersection)**.** *Assume assumption 7.1. For any component r and any nonempty set $J \subseteq \mathcal{N} \setminus \{r\}$,*

$$\bigcap_{a \in J} \mathrm{St}(V_{ra}) \neq \varnothing. \tag{7.8}$$

Proof. Subtrees of a tree satisfy the Helly property, so it is enough to prove pairwise intersection. If $\mathrm{St}(V_{ra})$ and $\mathrm{St}(V_{rb})$ were disjoint, an edge would separate them. On one side of that edge, r would agree with b; on the other, it would agree with a. The three distinct components a, r, b would violate (7.2). □

7.3 No unintended recombinants

The following statement is the key combinatorial consequence of the star condition.

Lemma 7.5 (Completion of a compatible branch)**.** *Orient an edge $e = (p, u)$ and let $T_{u|e}$ be the side containing u. Fix a block $\beta \in \mathcal{B}_e$. Choose one local type at every vertex of $T_{u|e}$ so that all internal bond labels agree and the boundary label on e is β. Then there is a target component $a \in \beta$ whose physical string on $T_{u|e}$ equals the chosen physical labels.*

Proof. We induct on $|T_{u|e}|$. The one-vertex case is immediate. For the induction step, let the chosen type at u be $\tau_u(c)$ for some $c \in \beta$. Let $f_1, \dots, f_m$ be the edges from u into the child subtrees $T_1, \dots, T_m$, and put $\gamma_i = \beta_{f_i}(c)$. By induction, the compatible assignment on T_i is the restriction of some component $c_i \in \gamma_i$.

Because c and c_i lie in the same star block across f_i, they agree on one side of that cut. If they agree on T_i, then c already matches the chosen assignment on that branch. Otherwise they agree on the complement of T_i, so c_i agrees with c at u and on every other branch.

There can be at most one index i of the second kind with $c_i \neq c$. If there were two, say $i \neq j$, then across the cut f_i the component c would agree with c_j on T_i and with c_i on the complement of T_i. The three distinct components c_j, c, c_i would violate (7.2).

If no such index exists, choose $a = c$. If exactly one exists, say i_0, choose $a = c_{i_0}$. This component matches the chosen assignment on T_{i_0} and agrees with c everywhere else, where c matches the remaining assignments. Finally, a and c agree on the side of the parent edge outside $T_{u|e}$, so they belong to the same block β. Thus $a \in \beta$. □

Theorem 7.6 (No unintended contraction terms)**.** *Under assumption 7.1, every compatible choice of one allowed local type at every vertex of $\mathcal{M}_\star$ produces exactly one target string. More precisely, if a contraction term has physical output x_a, then the chosen local type at every vertex is $\tau_v(a)$. Consequently,*

$$W(\theta) = \sum_{a \in \mathcal{N}} P_a(\theta) U_a, \qquad P_a(\theta) = \prod_{v \in V} g_{v,\tau_v(a)} \tag{7.9}$$

for every $\theta \in \mathcal{M}_\star$.

Proof. Root the tree. Apply the argument of lemma 7.5 to the child subtrees of the root. The same "at most one nontrivial child" argument shows that the physical labels of any globally compatible assignment coincide with one target string x_a.

It remains to identify the bond labels. Fix an edge e whose chosen label is β. Apply lemma 7.5 on each side. There are components in β whose restrictions equal x_a on the two respective sides. The edge of Γ_e labelled by a shares one endpoint with each of these edges, and therefore lies in the same connected component β. Hence $\beta_e(a) = \beta$. At every vertex, the chosen physical label and all incident block labels are therefore exactly $\tau_v(a)$. Since the target strings are distinct, a is unique, and the amplitude of U_a is the product in (7.9). □

7.4 Exact invariant dynamics

Theorem 7.7 (Exact block-aligned dynamics)**.** *Assume assumption 7.1 and $\chi_e \geq |\mathcal{B}_e|$ for every edge. Then $\mathcal{M}_\star$ is invariant under Euclidean gradient flow. For a solution with positive allowed entries, define auxiliary variables $z_a(t)$ by*

$$\dot{z}_a = 2(s_a - P_a)P_a, \qquad z_a(0) = 0. \tag{7.10}$$

Then

$$g_{v,\tau_v(a)}(t)^2 = g_{v,\tau_v(a)}(0)^2 + \sum_{b:\tau_v(b)=\tau_v(a)} z_b(t), \tag{7.11}$$

and the target amplitudes are given exactly by

$$P_a(t) = \prod_{v\in V} \left(g_{v,\tau_v(a)}(0)^2 + \sum_{b:\tau_v(b)=\tau_v(a)} z_b(t) \right)^{1/2}. \tag{7.12}$$

Proof. By theorem 7.6, the output on $\mathcal{M}_\star$ lies in the target span. Consider a forbidden coordinate at a vertex v. If it uses an unused bond label, its derivative output vanishes on $\mathcal{M}_\star$. Otherwise it specifies a physical label i and one block β_e on each incident edge. Each branch environment is a linear combination of branch restrictions of target strings belonging to the corresponding block, by lemma 7.5. If the derivative output had nonzero overlap with U_a, then $i = x_a(v)$ and, on every incident edge, the edge labelled by a would lie in β_e. The forbidden coordinate would then be precisely the allowed local type $\tau_v(a)$, a contradiction. Thus every transverse derivative is orthogonal to the target span. Invariance follows from theorem 6.1.

On $\mathcal{M}_\star$, theorem 7.6 gives

$$L = \frac{1}{2}\sum_a (P_a - s_a)^2. \tag{7.13}$$

For an allowed entry,

$$\frac{\partial L}{\partial g_{v,\tau}} = \sum_{a:\tau_v(a)=\tau} (P_a - s_a)\frac{P_a}{g_{v,\tau}}. \tag{7.14}$$

Therefore

$$\frac{\mathrm{d}}{\mathrm{d}t} g_{v,\tau}^2 = 2 \sum_{a:\tau_v(a)=\tau} (s_a - P_a)P_a = \sum_{a:\tau_v(a)=\tau} \dot{z}_a. \tag{7.15}$$

Integrating gives (7.11), and substitution into (7.9) gives (7.12). □

The positive orthant of allowed entries is forward invariant: at a boundary entry $g_{v,\tau} = 0$, every amplitude using that entry vanishes, and the vector field points inward or is tangent. The square-root representation is therefore valid for positive initialization.

Differentiating the amplitudes directly gives the equivalent kernel form

$$\dot{P} = \mathcal{K}(g)(s - P), \tag{7.16}$$

where

$$\mathcal{K}_{ab}(g) = P_a P_b \sum_{v:\tau_v(a)=\tau_v(b)} \frac{1}{g_{v,\tau_v(a)}^2}. \tag{7.17}$$

By lemma 7.3, the sum is over vertices outside $\mathrm{St}(V_{ab})$. Thus the tree determines the coupling kernel exactly.

Corollary 7.8 (Decoupled Saxe modes)**.** *Suppose* $\mathrm{St}(V_{ab}) = V$ *for every* $a \neq b$*, so no two components share a local entry. With uniform positive initialization* $g_{v,\tau_v(a)}(0) = \sigma$*, the amplitudes evolve independently and satisfy*

$$\dot{P}_a = K P_a^{2-2/K}(s_a - P_a). \tag{7.18}$$

Proof. No sharing gives

$$P_a = (\sigma^2 + z_a)^{K/2}. \tag{7.19}$$

Differentiate and use (7.10). □

In the general star-support case, the same local entry may carry several component amplitudes. The reduced system remains exact, but its modes interact through the shared factors in (7.12). This is the TTN-specific extension of aligned deep-linear dynamics.

8 Later saddles under the edgewise star condition

We now use the block-aligned representation to analyze a point at which several target components have been learned. The resulting connected route can combine different learned components on different branches; no unique parent is required.

8.1 Canonical partial-target points

Let $\varnothing \neq J \subsetneq \mathcal{N}$ be the learned set. At vertex v, call a local type active if it is used by at least one component in J:

$$\mathcal{T}_v(J) := \{\tau_v(a) : a \in J\}. \tag{8.1}$$

A block-aligned partial-target point has positive entries on $\mathcal{T}_v(J)$ and zero entries on the remaining allowed types.

Lemma 8.1 (Existence of a partial-target representative)**.** *Under assumption 7.1, there is a block-aligned parameter point θ_J such that*

$$W(\theta_J) = \sum_{a \in J} s_a U_a, \tag{8.2}$$

with precisely the local types in $\mathcal{T}_v(J)$ active.

Proof. For every $a \in J$, choose

$$v_a \in \bigcap_{b \in J \setminus \{a\}} \mathrm{St}(V_{ab}), \tag{8.3}$$

using corollary 7.4; if $J = \{a\}$, choose any vertex. By lemma 7.3, $\tau_{v_a}(a)$ is not used by any other learned component. Set every active entry equal to 1, then multiply the private entry $g_{v_a,\tau_{v_a}(a)}$ by s_a. These adjustments affect distinct learned amplitudes, so $P_a = s_a$ for all $a \in J$.

For a missing component r, corollary 7.4 gives a vertex at which its local type is not shared with any learned component; that entry is zero. Hence $P_r = 0$. Equation (8.2) follows from theorem 7.6. □

The representative is not unique, and no balancing assumption is needed for the order statements below.

8.2 The connected routing region

For $r \notin J$, define

$$\mathcal{R}_J(r) := \bigcap_{a \in J} \mathrm{St}(V_{ra}), \qquad c_J(r) := |\mathcal{R}_J(r)|, \qquad c_J^\star := \min_{r \notin J} c_J(r). \tag{8.4}$$

By corollary 7.4, $\mathcal{R}_J(r)$ is a nonempty connected subtree.

Proposition 8.2 (Dormant-entry characterization)**.** *A vertex v lies in $\mathcal{R}_J(r)$ if and only if the local type $\tau_v(r)$ is not used by any learned component. Equivalently,*

$$\mathcal{R}_J(r) = \{v : \tau_v(r) \notin \mathcal{T}_v(J)\}. \tag{8.5}$$

Proof. By lemma 7.3, $\tau_v(r) = \tau_v(a)$ exactly when $v \notin \mathrm{St}(V_{ra})$. Therefore no learned component shares $\tau_v(r)$ exactly when v belongs to every $\mathrm{St}(V_{ra})$. □

Thus $c_J(r)$ is the number of factors in the amplitude P_r that vanish at θ_J.

There is also an intrinsic branchwise characterization.

Proposition 8.3 (Minimal connected routing set)**.** *The subtree $\mathcal{R}_J(r)$ is the unique smallest connected set $C \subseteq V$ with the following property: on every connected component B of $T \setminus C$, the string $x_r|_B$ agrees with the restriction of at least one learned string.*

Proof. Let $U = \mathcal{R}_J(r)$ and let B be a component of $T \setminus U$. If every subtree $\mathrm{St}(V_{ra})$ met B, then, because each contains U and is connected, every one would contain the first vertex of B adjacent to U. That vertex would belong to their intersection, a contradiction. Hence B misses $\mathrm{St}(V_{ra})$ for some $a \in J$, and therefore $x_r = x_a$ on B.

Conversely, let C be connected and have the stated property. If $v \in U \setminus C$, let B be the component of $T \setminus C$ containing v. By assumption, $x_r = x_a$ on B for some $a \in J$. Hence V_{ra} lies in the connected complement of B, so $v \notin \mathrm{St}(V_{ra})$, contradicting $v \in U$. Therefore $U \subseteq C$. □

The proposition explains the geometry. Outside $\mathcal{R}_J(r)$, different branches of r may be supplied by different learned components. The entries inside $\mathcal{R}_J(r)$ are the irreducible connected graft needed to join those inherited branches.

8.3 Criticality and local order

Define the residual at θ_J by

$$R_J := W^\star - W(\theta_J) = \sum_{r \notin J} s_r U_r. \tag{8.6}$$

Theorem 8.4 (Structured later saddles)**.** *Assume assumption 7.1, and let θ_J be a partial-target point from lemma 8.1.*

1. *The point θ_J is critical in the full TTN parameter space if and only if*

$$c_J(r) \geq 2 \qquad \text{for every } r \notin J. \tag{8.7}$$

2. *If θ_J is critical, then*

$$\mathrm{ord}(\theta_J) \leq c_J^\star. \tag{8.8}$$

3. *Its restricted order on the invariant block-aligned manifold is*

$$\boxed{\mathrm{ord}_{\mathcal{M}_\star}(\theta_J) = c_J^\star.} \tag{8.9}$$

Proof. First consider tangent directions inside $\mathcal{M}_\star$. On this manifold,

$$L - L_J = \frac{1}{2} \sum_{a \in J} (P_a - s_a)^2 + \sum_{r \notin J} \left(\frac{1}{2} P_r^2 - s_r P_r \right). \tag{8.10}$$

At θ_J, the amplitude P_r is a monomial with exactly $c_J(r)$ zero factors, by proposition 8.2. Its first derivative is therefore nonzero exactly when $c_J(r) = 1$.

A coordinate transverse to $\mathcal{M}_\star$ has derivative output orthogonal to every target component by the proof of theorem 7.7. Hence it cannot contribute to the first derivative of the loss. This proves the criticality criterion.

Let $\varepsilon = \|\theta - \theta_J\|$ and restrict to $\mathcal{M}_\star$. Active entries remain bounded, whereas every dormant entry is $O(\varepsilon)$. Therefore

$$|P_r| \leq C \varepsilon^{c_J(r)}. \tag{8.11}$$

Dropping the nonnegative learned-component terms and using (8.10) gives

$$L - L_J \geq -C \varepsilon^{c_J^\star}. \tag{8.12}$$

Choose r with $c_J(r) = c_J^\star$ and turn on, with a common positive size τ, the dormant entries $g_{v,\tau_v(r)}$ for $v \in \mathcal{R}_J(r)$, leaving all other entries fixed. The learned amplitudes do not change because no learned component uses these entries. The amplitude of r is

$$P_r = A_r \tau^{c_J^\star}, \qquad A_r > 0. \tag{8.13}$$

Every other missing amplitude produced by the same perturbation is nonnegative and has degree at least $c_J^\star$; its linear contribution to the loss is therefore also nonpositive, while all norm-squared terms are of at least twice that degree. Hence

$$L - L_J \le -s_r A_r \tau^{c_J^\star} + O(\tau^{2c_J^\star}). \tag{8.14}$$

Together with (8.12), this proves (8.9). The same curve lies in the full parameter space, giving (8.8). □

Remark 8.5 (Why the full-space result is only an upper bound). The invariant manifold contains only target-supported routes. A transverse perturbation can first create non-target partial tensors, paying a positive quadratic cost, and two such changes may then combine into the residual. This is the spectral mechanism of theorem 3.2. The star condition removes unintended recombinants on $\mathcal{M}_\star$; it does not remove every transverse spectral shortcut.

Corollary 8.6 (Monotonicity of connected route depth). *If $J \subseteq J'$, then*

$$\mathcal{R}_{J'}(r) \subseteq \mathcal{R}_J(r), \qquad c_{J'}(r) \le c_J(r). \tag{8.15}$$

Thus learning additional components can only shorten the connected route of a remaining component.

Example 8.7 (Branchwise splicing). On a seven-vertex path, suppose the learned components are

$$0000000, \qquad 1111111, \tag{8.16}$$

and the missing component is

$$0022111. \tag{8.17}$$

Relative to the first learned string, the disagreement subtree is the interval from vertices 2 through 6; relative to the second, it is the interval from vertices 0 through 3. Their intersection is the two-vertex interval $\{2, 3\}$. Hence $c_J(r) = 2$, although budding from either learned component alone would cost 5 or 4 vertices. The missing component reuses the first learned route on the left and the second on the right, joined by a two-vertex connected graft. By theorem 8.4, the partial-target point is a strict saddle.

The exact reduced equations in theorem 7.7 now have a natural interpretation. At the partial-target point, $c_J(r)$ is the number of zero local factors in P_r. In an isolated balanced route, it is therefore the effective depth d in (6.15). Unlike architectural depth, it changes with J.

9 An exact full-space theorem for block-separated later saddles

The star-support theorem gives exact multi-component dynamics and an exact restricted order, but it does not exclude transverse spectral shortcuts. A complementary full-space result is available when residual families and perturbation channels separate into orthogonal parent blocks. The assumptions are stronger, but the conclusion includes both direct and spectral mechanisms exactly.

Let J be the learned set and assign each residual component $r \notin J$ a parent $\pi(r) \in J$. Define

$$R^{(a)} := \sum_{r:\pi(r)=a} s_r U_r, \qquad J_+ := \left\{ a \in J : R^{(a)} \neq 0 \right\}. \tag{9.1}$$

For $a \in J_+$, define

$$V_{r|a} := \{ v : x_r(v) \neq x_a(v) \}, \qquad k_a^\star := \min_{r:\pi(r)=a} |\operatorname{St}(V_{r|a})|. \tag{9.2}$$

For each $a \in J_+$, repeat the one-component atomic construction relative to U_a, using only the residual $R^{(a)}$. Let $M_{a,\ell}$ be the resulting coupling matrix,

$$\mu_{a,\ell} := \lambda_{\max}(M_{a,\ell}), \qquad \ell_a^\star := \min \{ \ell : \mu_{a,\ell} > s_a \}. \tag{9.3}$$

We use the following explicit separation hypothesis.

Assumption 9.1 (Orthogonal parent-block decomposition)**.** Near the candidate saddle, after removing gauge directions, the perturbation variables split into blocks $u^{(a)}$, $a \in J_+$, and stable mixed variables such that

$$L - L_S = \sum_{a \in J_+} \Delta L_a(u^{(a)}) + Q_{\mathrm{mix}}(u), \tag{9.4}$$

where:

1. ΔL_a is exactly the one-component loss problem with learned component $s_a U_a$ and residual $R^{(a)}$;
2. $Q_{\mathrm{mix}}(u) \geq 0$ near the saddle;
3. each parent block has enough private channels for the relevant direct and spectral constructions;
4. every parent block is critical and satisfies the nonboundary hypotheses of theorem 3.2.

The assumption is satisfied, for example, when learned components occupy orthogonal component channels, each residual family uses fresh channels reserved for one parent, and every mixed contraction involving two parent families is orthogonal to the target residual and to the leading single-parent sectors.

Theorem 9.2 (Block-separated later-saddle order)**.** *Under assumption 9.1,*

$$\boxed{\operatorname{ord}(\theta_S) = \min_{a \in J_+} \min\{k_a^\star, 2\ell_a^\star\}.} \tag{9.5}$$

Proof. For each parent, theorem 3.2 gives

$$\Delta L_a(u^{(a)}) \geq -C_a \left\| u^{(a)} \right\|^{p_a}, \qquad p_a := \min\{k_a^\star, 2\ell_a^\star\}. \tag{9.6}$$

Let $p = \min_a p_a$. For $\|u\| \leq 1$,

$$\left\| u^{(a)} \right\|^{p_a} \leq \|u\|^p. \tag{9.7}$$

Using $Q_{\mathrm{mix}} \geq 0$ in (9.4) gives

$$L - L_S \geq -C \|u\|^p. \tag{9.8}$$

Choose a parent with $p_a = p$. The direct or spectral construction inside its private block, with all other variables zero, gives a curve with loss decrease $-c\tau^p + o(\tau^p)$. Hence the upper and lower orders agree. □

This theorem is deliberately conservative. It is a genuine later-saddle full-space result, but it excludes the branchwise channel splicing captured by section 8. The unrestricted problem must combine both phenomena: state-dependent connected routing and transverse cooperative instabilities.

10 Synthesis and scope

The results above give a coherent local picture of TTN training.

10.1 Two mechanisms

At a one-component saddle, every loss-decreasing perturbation is controlled by one of two mechanisms.

1. A *connected direct route* creates a residual component as one irreducible output change. Its order is the number of local factors forced by tree connectivity.
2. A *spectral route* combines cheaper partial changes. The individual changes pay positive quadratic cost, while compatible pairs gain residual correlation. A generalized eigenvalue crossing determines when the gain wins.

The exact formula (3.18) says that the faster of these mechanisms determines the saddle order. The connected cumulant coordinates are what make this dichotomy complete rather than merely constructive.

At a structured later saddle, the connected direct route is no longer measured relative to one parent. Under the edgewise star condition it is the intersection

$$\mathcal{R}_J(r) = \bigcap_{a \in J} \mathrm{St}(V_{ra}). \tag{10.1}$$

This is the portion of the route that cannot be inherited from any learned component. Different branches outside $\mathcal{R}_J(r)$ may come from different components. The route depth is therefore a property of the current learned set, not only of the missing target.

10.2 The precise analogue of aligned deep-linear dynamics

The comparison with deep linear networks can be summarized as follows.

	Aligned deep linear network	Tree tensor network
Basic mode	singular direction	compatible target route
Invariant coordinates	singular-mode amplitudes	local route entries or block-aligned types
Amplitude kernel	diagonal	JJ^{T}; off-diagonal through shared entries
Depth	fixed number of layers	number of currently dormant route factors
Balance law	differences of layer Gram matrices	route-square invariants and bond Gram balance
Local scalar equation	$\dot{y} = Ly^{2-2/L}(s-y)$	$\dot{y} = d\gamma^{2/d}y^{2-2/d}(s-y)$
Later learning	next singular mode	route structure changes with learned set J

The exact block-aligned equations strengthen this analogy. They do not merely assign an effective depth to each component; they specify how components interact through shared local factors. The relation

$$\tau_v(a) = \tau_v(b) \quad \Longleftrightarrow \quad v \notin \mathrm{St}(V_{ab}) \tag{10.2}$$

turns parameter sharing into a geometric statement about the target support.

10.3 What is proved

The following statements are complete under their stated assumptions.

1. For a one-component saddle, theorem 3.2 gives the exact full-space order $\min\{k^\star, 2\ell^\star\}$.
2. On any routing-closed manifold satisfying transverse orthogonality, theorem 6.1 gives exact amplitude dynamics.
3. Under the edgewise star condition, theorems 7.6 and 7.7 construct an explicit invariant manifold and exact coupled scalar reduction.
4. For a canonical partial-target point under that condition, theorem 8.4 gives the full-space criticality criterion, a full-space direct upper bound, and the exact restricted order.
5. Under orthogonal parent-block separation, theorem 9.2 gives an exact full-space later-saddle order including parentwise spectral shortcuts.

10.4 What is not proved

Three logically distinct problems remain.

First, for an arbitrary aligned multi-channel saddle, there is not yet a general connected-coordinate theorem that rules out unenumerated lower-order combinations of channel transitions. The expected direct-versus-spectral formula is stated conditionally in section B.

Second, a local order theorem does not prove that generic small random initialization reaches a particular saddle representative. Even on an invariant manifold, a complete saddle-to-saddle theorem requires uniform control of coupled routes, ties, and the amplitudes accumulated before each plateau. Deep linear networks already require additional arguments of this kind [2].

Third, invariance is weaker than attraction. The block-aligned manifold is exactly invariant under the edgewise star condition, but the sign of transverse quadratic terms determines whether nearby generic trajectories are drawn toward it or leave it. This is a stability question, not a consequence of the invariance proof.

10.5 Conclusion

The central conclusion is that TTN training has a natural route geometry. A missing component is not characterized only by its coefficient or by its distance from one learned component. It is characterized by the local entries that can be inherited from the current active network, the connected region that must still be created, and the possibility that cheaper partial changes cooperate off that route.

For one learned component this geometry is exact and complete:

$$\text{saddle order} = \min\{\text{connected Steiner order}, \text{spectral pair order}\}. \tag{10.3}$$

For structured later saddles, the connected Steiner order becomes the state-dependent intersection cost $|\mathcal{R}_J(r)|$, and the aligned dynamics become exact coupled Saxe equations. The remaining general problem is sharply isolated: construct multi-channel connected coordinates strong enough to prove that the same direct-versus-spectral dichotomy persists without the edgewise star or parent-separation assumptions.

A Detailed bookkeeping for the filtered lower bound

We expand the estimate used in proposition 4.3. Fix ℓ and put

$$p := \min(k^\star, 2\ell). \tag{A.1}$$

A nonempty set A is small when $|\operatorname{St}(A)| \le \ell - 1$ and large otherwise. Let $\kappa_s = (\kappa_A)_{A \text{ small}}$.

For a small B, every block in a partition of B is small, and

$$m_B = \kappa_B + \sum_{\substack{\pi \in \operatorname{Part}(B) \\ |\pi| \ge 2}} \bigotimes_{A \in \pi} \kappa_A. \tag{A.2}$$

All nonempty connected coordinates are $O(\varepsilon)$. Hence the remainder is bounded by $C\varepsilon \|\kappa_s\|$ after summing over the finitely many small sectors. Squaring and summing gives

$$\sum_{B \text{ small}} \|m_B\|^2 \ge (1 - C\varepsilon) \|\kappa_s\|^2 . \tag{A.3}$$

Dropping the nonnegative terms with large B gives (4.26).

Now consider the residual reward. Only sectors with $R_B \neq 0$ matter, and every such sector satisfies $|\operatorname{St}(B)| \ge k^\star$. Expanding m_B by partitions gives the following cases.

One block. The contribution is κ_B. By lemma 4.2,

$$\|\kappa_B\| = O(\varepsilon^{|\operatorname{St}(B)|}) = O(\varepsilon^{k^\star}). \tag{A.4}$$

Two small blocks. The unordered partitions $B = A_1 \sqcup A_2$ with both blocks small contribute

$$\frac{1}{2} Q(\kappa_s, \kappa_s) \le \frac{1}{2} \lambda_{\max}(S_{\ell-1}) \|\kappa_s\|^2 . \tag{A.5}$$

The factor $1/2$ compensates for the ordered sum in (4.22).

One small and one large block. If A is small and C is large, then $\|\kappa_C\| = O(\varepsilon^\ell)$. For every $\gamma > 0$, Young's inequality gives

$$\|\kappa_A\| \|\kappa_C\| \le \frac{\gamma}{2} \|\kappa_A\|^2 + C_\gamma \varepsilon^{2\ell}. \tag{A.6}$$

After summation over finitely many partitions, the first term contributes $M\gamma \|\kappa_s\|^2$.

Two or more large blocks. Two large blocks contribute $O(\varepsilon^{2\ell})$. Any additional factors only improve the order.

At least three blocks with two small blocks. Select two small blocks. Every remaining connected coordinate is $O(\varepsilon)$, so the term is bounded by

$$C\varepsilon \left(\|\kappa_{A_1}\|^2 + \|\kappa_{A_2}\|^2 \right). \tag{A.7}$$

Summing gives $C\varepsilon \|\kappa_s\|^2$.

Combining the cases yields (4.27). In the master identity, the coefficient of $\|\kappa_s\|^2$ is

$$\frac{\omega}{2} \left[\omega(1 - C\varepsilon) - \lambda_{\max}(S_{\ell-1}) - M\gamma - C\varepsilon \right]. \tag{A.8}$$

Since $\omega = s_j + O(\varepsilon)$ and $s_j - \lambda_{\max}(S_{\ell-1}) > 0$, it is positive after first choosing γ and then ε small enough. The remaining negative terms are $O(\varepsilon^{k^\star}) + O(\varepsilon^{2\ell}) = O(\varepsilon^p)$.

B Conditional theory for arbitrary aligned later saddles

This appendix formulates the expected extension beyond the two proved later-saddle classes. The result is conditional on a multi-channel connected-expansion hypothesis. Nothing in this appendix is used in the proofs of the main text.

B.1 Why a raw defect count is insufficient

At an aligned later saddle, let $\mathcal{A}_v$ be the set of active local core entries at vertex v. A contraction monomial producing a missing product component can be assigned a *raw defect count*: the number of vertices at which it uses an entry outside $\mathcal{A}_v$. This count is useful for enumerating Taylor monomials, but it is not the order of a direct descent.

Example B.1 (A low-degree monomial that is not a descent). Consider a three-vertex path, learned tensor $U_0 = 000$ with coefficient $s_0 = 1$, and one residual component $U_r = 101$ with $0 < s_r < 1$. Keep the active channel and add physical rotations x and z at the two endpoints. Then

$$W = 000 + x\,100 + z\,001 + xz\,101. \tag{B.1}$$

The residual-producing monomial xz uses only two new entries, but

$$L - L_S = \frac{1}{2}(x^2 + z^2 + x^2z^2) - s_r xz. \tag{B.2}$$

The quadratic Hessian in (x, z) has eigenvalues $1 \pm s_r$ and is positive definite. Thus the two-entry monomial is not a loss-decreasing direction. The direct all-or-none construction uses a private channel through all three vertices and has order three, in agreement with corollary 4.4. If $s_r > 1$, the quadratic direction does become unstable; it is then a spectral shortcut, not a connected direct route.

The correct direct quantity must exclude lower-order off-target outputs. Equivalently, it must measure an irreducible connected output change, not merely one monomial within a larger contraction expansion.

B.2 Connected routed coordinates

Let θ_S be an aligned multi-channel saddle with residual

$$R = W^\star - W(\theta_S). \tag{B.3}$$

The following is the missing structural statement.

Assumption B.2 (Multi-channel connected expansion)**.** There is a finite family of analytic connected coordinates κ_α near θ_S, each with an integer cost $d(\alpha) \geq 1$ and a leading output tensor X_α, with the following properties.

1. Every normalized output coefficient is a finite sum of products of compatible connected coordinates, analogously to a moment–cumulant expansion.
2. If $\varepsilon = \|\theta - \theta_S\|$, then
$$\|\kappa_\alpha\| \leq C\varepsilon^{d(\alpha)}. \tag{B.4}$$
More strongly, κ_α vanishes whenever any local defect block required by the routed pattern α is set to zero.
3. The one-block contribution of κ_α to the output is
$$\kappa_\alpha X_\alpha + \text{higher-order terms}. \tag{B.5}$$
4. Products of two connected coordinates generate the complete leading pairwise residual coupling; products of three or more have an additional small factor unless at least two factors lie above the current filtration level.

For the one-component saddle, the index α is a nonempty vertex set with a physical tensor label, $d(\alpha) = |\operatorname{St}(\alpha)|$, and assumption B.2 is exactly the cumulant construction and subtree-divisibility lemma proved in section 4.

Define the connected direct degree

$$d^\star := \min\{d(\alpha) : \langle R, X_\alpha\rangle \neq 0\}. \tag{B.6}$$

A connected coordinate attaining $d^\star$ gives a direct loss decrease of order $d^\star$, provided it can be realized with the stated local cost.

An equivalent intrinsic formulation is the smallest integer d for which there is an analytic arc $\theta(t)$ satisfying

$$W(\theta(t)) - W(\theta_S) = t^d X + O(t^{d+1}), \qquad \langle R, X\rangle > 0. \tag{B.7}$$

The absence of lower-order output terms is what distinguishes (B.7) from a raw contraction monomial.

B.3 Filtered spectral coupling

For $\ell \geq 1$, let $\mathcal{X}_\ell$ be the finite-dimensional space generated by connected atoms of cost at most ℓ that have no direct residual overlap at the current filtration level. After quotienting gauge directions and minimizing purely stable variables, write the output expansion in atom amplitudes $x \in \mathcal{X}_\ell$ as

$$\Delta W = X_\ell x + \frac{1}{2} B_\ell(x, x) + O(\|x\|^3). \tag{B.8}$$

Define the positive Gram form

$$A_\ell := X_\ell^* X_\ell \tag{B.9}$$

and the residual coupling form C_ℓ by

$$\langle x, C_\ell x\rangle := \langle R, B_\ell(x, x)\rangle. \tag{B.10}$$

Assume A_ℓ is positive definite on the gauge quotient. The leading atom energy is

$$\Delta L = \frac{1}{2}\langle x, (A_\ell - C_\ell)x\rangle + O(\|x\|^3). \tag{B.11}$$

Set

$$\Lambda_\ell := \lambda_{\max}\left(A_\ell^{-1/2} C_\ell A_\ell^{-1/2}\right), \qquad \ell^\star := \min\{\ell : \Lambda_\ell > 1\}. \tag{B.12}$$

At the one-component saddle, $A_\ell = s_j^2 \mathrm{I}$ and $C_\ell = s_j M_\ell$, so $\Lambda_\ell = \mu_\ell / s_j$.

The spectral upper construction also requires realizability: a generalized eigenvector must be implementable by compatible local perturbations whose parameter scale is $O(\eta^{1/\ell})$, without unintended lower-order contractions. Private channels are a sufficient, though not necessary, condition.

B.4 Conditional order theorem

Theorem B.3 (Conditional general later-saddle formula)**.** *Assume:*

1. *the connected expansion in assumption B.2;*
2. *coercivity of A_ℓ on the gauge quotient;*
3. *strict nonboundary inequalities $\Lambda_\ell \neq 1$ below the relevant direct scale;*
4. *realizability of a connected atom attaining $d^\star$ and of the first unstable generalized eigenvector;*
5. *no accidental cancellation of the leading direct residual coefficient.*

Then

$$\boxed{\operatorname{ord}(\theta_S) = \min\{d^\star, 2\ell^\star\}.} \tag{B.13}$$

Proof. The realizable connected atom and generalized eigenvector give the two upper bounds

$$\operatorname{ord}(\theta_S) \le d^\star, \qquad \operatorname{ord}(\theta_S) \le 2\ell^\star. \tag{B.14}$$

For the lower bound, fix ℓ below the first spectral threshold. Call a connected coordinate cheap when $d(\alpha) \le \ell - 1$ and expensive otherwise. By (B.4), expensive coordinates are $O(\varepsilon^\ell)$. The positive output norm gives the coercive quadratic form $A_{\ell-1}$ on cheap coordinates. Pairwise residual terms among cheap coordinates are controlled by $C_{\ell-1}$, and $\Lambda_{\ell-1} < 1$ leaves a strict positive gap. A cheap–expensive product is absorbed by Young's inequality; two expensive factors are $O(\varepsilon^{2\ell})$; and any product of at least three connected factors either contains two expensive factors or carries an additional $O(\varepsilon)$ multiplying the cheap quadratic form. The first one-block residual term occurs at order $d^\star$. Therefore

$$L - L_S \ge -C\varepsilon^{\min(d^\star, 2\ell)}. \tag{B.15}$$

Choosing ℓ exactly as in the proof of theorem 3.2 gives the lower bound in (B.13). □

The theorem shows that once the correct connected coordinates exist, the remainder of the one-component proof is stable under considerable generalization. The open problem is structural, not spectral.

B.5 Possible routes to the missing theorem

For an aligned matrix product state, contraction monomials are walks in a finite channel-state graph. A connected defect is an interval word of matrix units. This suggests ordered cumulants or a one-dimensional polymer expansion in channel-transition defects.

For a general tree, removing the defect vertices partitions the network into regions carrying active learned labels. The natural objects are connected defect polymers with boundary channel labels. Three equivalent-looking formulations may be useful:

1. channel-conditioned cumulants;
2. a linked-cluster expansion over routed defect polymers;
3. the associated graded algebra of the TTN contraction map under the filtration that gives active entries weight zero and novel entries weight one.

The last formulation makes clear why raw monomial degree is insufficient: the initial form records every cheap output, whereas the connected primitive part is needed to separate direct residual production from products of cheaper changes.

C Formal small-initialization consequences

The exact reduced system (7.12) suggests a simple sequential-learning picture, but a complete theorem would require uniform control of the trajectory between successive saddle neighbourhoods. This appendix records the consequences under an explicit time-scale-separation hypothesis; they are not used elsewhere.

C.1 Spectral-route normal forms

The one-component order theorem identifies a spectral instability of order 2ℓ when two atoms of cost at most ℓ cooperate. A corresponding scalar dynamical law requires a reduction to the unstable normal form. Precisely, suppose that after the stable atom variables have been

minimized, a balanced core coordinate b parametrizes the unstable direction and the reduced loss satisfies

$$L - L_S = -cb^{2\ell} + o(b^{2\ell}), \qquad c > 0. \tag{C.1}$$

Then

$$\dot{b} = 2\ell c\, b^{2\ell-1} + o(b^{2\ell-1}), \tag{C.2}$$

and the escape scale is the scalar-normal-form scale associated with order 2ℓ. This is a conditional normal-form statement; the local order theorem alone does not provide the required dynamical reduction.

C.2 Connected-route plateaus

Suppose the flow is near a partial-target point θ_J and consider a missing component r. Write

$$c = c_J(r), \qquad A_r(J) := \prod_{v \notin \mathcal{R}_J(r)} g^J_{v,\tau_v(r)}, \tag{C.3}$$

where g^J denotes the active entries at the saddle. If the c dormant entries of r are isolated from competing missing routes and are balanced at a common value b, then locally

$$P_r = A_r(J) b^c + o(b^c), \tag{C.4}$$

$$\dot{b} = s_r A_r(J) b^{c-1} + o(b^{c-1}). \tag{C.5}$$

Consequently,

$$t_r(J) \asymp \begin{cases} \big(s_r A_r(J)\big)^{-1} \log(1/\sigma), & c = 2, \\ \big((c-2) s_r A_r(J)\big)^{-1} \sigma^{-(c-2)}, & c \geq 3. \end{cases} \tag{C.6}$$

The power of σ depends only on the connected route depth; the coefficient $s_r A_r(J)$ breaks ties between equal depths.

Assumption C.1 (Separated route time scales)**.** Along a sequence of partial-target neighbourhoods:

1. the dormant entries of every candidate route are of a common order σ upon entry;
2. active factors remain bounded above and below independently of σ;
3. the leading candidate routes are dynamically isolated up to lower-order corrections;
4. after one candidate reaches order one, the flow enters the next partial-target neighbourhood before any higher-cost route changes by order one.

Under this assumption, (C.6) immediately implies a greedy ordering rule: first minimize $c_J(r)$, then compare the coefficient $s_r A_r(J)$ among ties. The statement is asymptotic in the common initialization scale. It should not be confused with a proof that generic full-space training satisfies assumption C.1.

A useful special case has a familiar combinatorial interpretation. Define the pairwise Steiner cost

$$d(a, r) := |\operatorname{St}(V_{ar})|. \tag{C.7}$$

Suppose that at every visited state J and for every missing r, the intersection $\mathcal{R}_J(r)$ is equal to one of the pairwise subtrees $\operatorname{St}(V_{ar})$, $a \in J$. Then

$$c_J(r) = \min_{a \in J} d(a, r). \tag{C.8}$$

The greedy route-depth rule is exactly Prim's algorithm on the complete graph of target components with edge weights $d(a, r)$. The resulting parent graph is a minimum spanning tree for these weights. This is a special nested-subtree regime; the function d need not satisfy the triangle inequality, and in the general splicing case the intersection cost is not a pairwise distance to any single learned component.

The formal picture can therefore be summarized as

$$J \longmapsto \{c_J(r), A_r(J)\}_{r \notin J} \longmapsto \text{next route} \longmapsto J \cup \{r\}. \tag{C.9}$$

The exact theorems of the main text justify the route depths and the reduced equations. What remains for a full saddle-to-saddle theorem is the dynamical control encoded in assumption C.1.

D Notation and logical dependencies

D.1 Principal notation

Symbol	Meaning
$T = (V, E)$	Tree underlying the TTN
G_v	Core tensor at vertex v
χ_e	Bond dimension on edge e
$W(\theta)$	Contracted tensor
$W^\star = \sum_a s_a U_a$	Orthogonal product-basis target
V_{ab}	Vertices where target strings a and b disagree
$\mathrm{St}(B)$	Smallest connected subtree containing B
θ_S	One-component saddle representative
R_B	Residual sector differing from U_j on exactly B
$k^\star$	Smallest relevant residual Steiner-subtree size
m_B	Normalized output sector $P_B W/\omega$
κ_B	Connected sector coordinate $\log_* m$
M_ℓ	Pairwise coupling matrix for atoms of cost at most ℓ
μ_ℓ	Largest eigenvalue of M_ℓ
$\ell^\star$	First level with $\mu_\ell > s_j$
Γ_e	Bipartite support graph across edge e
$\beta_e(a)$	Star block of component a across edge e
$\tau_v(a)$	Local physical and incident-block type of component a
$\mathcal{M}_\star$	Block-aligned invariant manifold
$\mathcal{R}_J(r)$	Connected routing region $\bigcap_{a \in J} \mathrm{St}(V_{ra})$
$c_J(r)$	Connected route depth $\lvert\mathcal{R}_J(r)\rvert$
$d^\star$	Conditional connected direct degree at a general aligned saddle
A_ℓ, C_ℓ	Conditional positive and residual-coupling forms
Λ_ℓ	Conditional generalized spectral eigenvalue

D.2 Dependency map

The exact one-component theorem depends on the following chain.

1. Orthogonal sectors give criticality and the master loss identity.
2. The finite cumulant transform separates connected from disconnected output changes.

3. Tree factorization implies subtree divisibility of the connected coordinates.
4. The strict spectral gap controls all cheap pairwise changes in the lower bound.
5. Direct and Perron-eigenvector constructions give matching upper bounds.

The exact block-aligned dynamics depend on a different chain.

1. The edgewise star condition makes every edge support block rank one.
2. The branch-completion lemma rules out unintended recombinant contractions.
3. Transverse derivatives are orthogonal to the target span, giving invariance.
4. The product formula for target amplitudes gives the exact auxiliary-variable reduction.
5. The sharing lemma identifies dormant route factors with intersections of Steiner subtrees.

The conditional theorem in section B would follow from a multi-channel replacement for the third item in the first chain: connected coordinates with a defect-divisibility estimate. No statement from that appendix is needed for any theorem in the main text.

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